

TELCO CUSTOMER CHURN – DATASET INSPECTION REPORT

Analyst Lab Africa Data Analytics Internship Programme

Week 1: Business Analytics Case Study

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Dataset: Telco Customer Churn Dataset

Dataset Overview

The **Telco Customer Churn Dataset** comprises **7,043 customer records and 21 variables**. Each row represents an individual customer and contains information relating to the customer's demographic characteristics, account details, subscribed services, payment methods, and churn status.

The dataset includes key variables such as **gender, senior citizen status, partner and dependent status, tenure, contract type, internet and other subscribed services, payment method, monthly charges, total charges, and churn status**. The **Churn** variable indicates whether a customer remained with the company or discontinued its services.

The dataset was used to examine customer behavior, identify patterns associated with churn, and generate insights that can support customer retention strategies.

Rows and Columns

The dataset contains **7,043 rows and 21 columns**. Each row represents an individual customer, while each column contains a specific attribute relating to the customer.

The 21 columns are grouped into the following categories:

- **Customer identification:** customerid
- **Demographic information:** gender, senior citizen, partner, dependents
- **Customer account information:** tenure, contract, paperless billing
- **Phone and internet services:** phone service, multiple lines, internet service
- **Additional services:** online security, online backup, device protection, Tech support, streaming, streaming movies

- **Payment information:** payment method
- **Financial information:** monthly charges, total charges
- **Churn information:** churn

These variables provide information about the customers' **demographics, length of service, subscribed services, contract arrangements, payment methods, charges, and churn status**, which are used to analyze customer behavior and identify factors associated with customer retention and churn.

Data Types

The variables in the dataset consist of different data types based on the type of information they contain. The main data types identified are **categorical, numerical, and identifier variables**.

- **Identifier:** customer_id is used to uniquely identify each customer.
- **Categorical variables:** gender, senior citizen, partner, dependents, phone service, multiple lines, internet service, online security, online backup, device protection, Tech support, streaming TV, streaming movies, contract, paperless billing, payment method, and churn.
- **Numerical variables:** tenure, monthly charges, and total charges.

These data types were reviewed to ensure that the variables were appropriately prepared for analysis, calculations, PivotTables, and visualizations.

Missing Values

A missing-value check was performed across all **21 variables** in the dataset. The inspection showed that **there were no missing or blank values** in the dataset. Therefore, no missing-value treatment or imputation was required before proceeding with the analysis.

The completeness of the dataset ensured that the available records could be used directly for the subsequent analysis and visualizations.

Duplicate Records

A duplicate-record check was performed to identify whether any customer records appeared more than once in the dataset. The inspection showed that no duplicate records were identified.

Therefore, no duplicate records were removed, and all 7,043 customer records were retained for the analysis.

Summary Statistics

Summary statistics were reviewed for the main numerical variables in the dataset, including **tenure**, **monthly charges**, and **total charges**. These variables provide an overview of customers' length of service and financial activity.

The **tenure** variable represents the number of months a customer has remained with the company, while **monthly charges** and **total charges** represent the amount charged to customers on a monthly and cumulative basis, respectively.

These summary statistics were used to understand the distribution of the numerical variables and support further analysis of customer behavior and churn.

Data Cleaning and Preparation

Several data preparation steps were carried out to ensure that the dataset was suitable for analysis and visualization.

- The **senior citizen** variable was originally recorded using numerical values, where **0 represented “No”** and **1 represented “Yes.”** These values were converted to **“No” and “Yes”** to make the variable easier to interpret.
- The dataset was checked for **missing values**, and no missing values were identified.
- The dataset was checked for **duplicate records**, and no duplicates were identified.
- The data types of the variables were reviewed to ensure that they were appropriate for analysis.
- The prepared dataset was then used to create **PivotTables, charts, and the customer churn dashboard**.

Conclusion

The Telco Customer Churn Dataset was inspected and prepared for analysis. The dataset contains 7,043 customer records and 21 variables, with no missing values or duplicate records identified during the inspection. The variables were reviewed and appropriately prepared for analysis and visualization. The resulting dataset provided a suitable foundation for examining customer churn patterns and developing business insights and recommendations.